

JEE MAIN REVISION MASTER: THREE DIMENSIONAL GEOMETRY

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 41 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Planes & Line-Plane Intersections in 3D	24 PYQs	3D Planes
Very High	Direction Cosines & Equations of Lines in 3D	18 PYQs	3D Lines

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Planes & Line-Plane Intersections in 3D	$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$, $d = \frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}$, $\sin\theta = \frac{ \vec{b} \cdot \vec{n} }{ \vec{b} \vec{n} }$
Direction Cosines & Equations of Lines in 3D	$l^2 + m^2 + n^2 = 1$, $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$, $d_{\text{skew}} = \frac{ \vec{a}_2 \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Planes & Line-Plane Intersections in 3D (Very High Yield — 24 PYQs)

Core Principle: Normal vector to a plane, perpendicular distance from point to plane, and angle between a line and plane (uses $\sin\theta$).

■ **Caution / JEE Pitfall:** Angle between a LINE (vector $\vec{b}$) and a PLANE (normal $\vec{n}$) is computed with SINE ($\sin\theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$), NOT cosine!

■ **Speed Trick:** A line is parallel to a plane if and only if it is perpendicular to the plane's normal: $\vec{b} \cdot \vec{n} = 0$.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): "The equation of the line through the point (0,1,2) and perpendicular to the line $(x-1)/2 = (y+1)/3 = (z-1)/-2$ is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The distance of the point P(-3,4,5) from yz plane is..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "The sine of the angle between the straight line $(x-2)/3 = (y-3)/4 = (z-4)/-5$ and the plane $2x-2y+z=5$ is..." [Ans: N/A]

3.2 Direction Cosines & Equations of Lines in 3D (Very High Yield — 18 PYQs)

Core Principle: Direction cosines (l, m, n), line passing through two points, and shortest distance formula between skew lines.

■ **Caution / JEE Pitfall:** Direction ratios (a, b, c) can be any proportional numbers, but direction cosines (l, m, n) MUST satisfy $l^2 + m^2 + n^2 = 1$.

■ **Speed Trick:** Two lines in 3D intersect (are coplanar) if and only if the shortest distance between them is ZERO: $(\text{vec}(a)_2 - \text{vec}(a)_1) \cdot (\text{vec}(b)_1 \times \text{vec}(b)_2) = 0$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The three points $A(2,4,3), B(4,a,9)$ and $C(10,-1,7)$ form a right-angled triangle with $\angle B = 90^\circ$, then the value of "a" is..." **[Ans: B]**
- **[2026 Shift] Q#2 (MCQ):** "The measure of the angle between the lines $x=k+1, y=2k-1, z=2k+3, k \in \mathbb{R}$ and $(x-1)/2 = (y-2)/1 = (z-3)/1$ is..." **[Ans: B]**
- **[2026 Shift] Q#3 (MCQ):** "The angle between the lines whose direction ratios are a, b, c and $b-c, c-a, a-b$ is..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The three points $A(2,4,3), B(4,a,9)$ and $C(10,-1,7)$ form a right-angled triangle with $\angle B = 90^\circ$, then the value of "a" is

- (A) 1 or 4
- (B) -1 or 4 **[Correct]**
- (C) 1 or -4
- (D) -1 or -4

Correct Answer: B

Solution Analysis: Since the triangle is right-angled at B, the two sides meeting at B must be perpendicular. So, vectors $\text{vec}(BA)$ and $\text{vec}(BC)$ must satisfy $\text{vec}(BA) \cdot \text{vec}(BC) = 0$. Now, $A(2,4,3), B(4,a,9), C(10,-1,7)$. So, $\text{vec}(BA) = A - B$
 $\text{vec}(BA) = (2-4, 4-a, 3-9) = (-2, 4-a, -6)$. Also, $\text{vec}(BC) = C - B$
 $\text{vec}(BC) = (10-4, -1-a, 7-9) = (6, -1-a, -2)$. Since they are perpendicular, $(-2)(6) + (4-a)(-1-a) + (-6)(-2) = 0$
 $-12 + (4-a)(-1-a) + 12 = 0$
 $(4-a)(-1-a) = 0$
 $4-a = 0$ or $-1-a = 0$
Hence, $a = 4$ or $a = -1$. So the correct option is Option B-1 or 4.

Benchmark Problem 2: KCET 2025 (Q#1)

The equation of the line through the point $(0,1,2)$ and perpendicular to the line $(x-1)/2 = (y+1)/3 = (z-1)/-2$ is

- (A) $(x/3) = (y-1)/4 = (z-2)/-3$
- (B) $(x/-3) = (y-1)/4 = (z-2)/3$
- (C) $(x/3) = (y-1)/4 = (z-2)/3$
- (D) $(x/3) = (y-1)/-4 = (z-2)/3$

Correct Answer: None

Solution Analysis: The given line has direction vector $v = (2, 3, -2)$. We seek a direction $d = (d_1, d_2, d_3)$ with $v \cdot d = 2d_1 + 3d_2 - 2d_3 = 0$. One convenient choice is $d_1 = -3, d_2 = 4, d_3 = 3$ since $2(-3) + 3(4) - 2(3) = -6 + 12 - 6 = 0$. Hence the required line through $(0,1,2)$ is $(x-0)/-3 = (y-1)/4 = (z-2)/3$, which is Option B.